Maynooth University
Department of Electronic Engineering
EE499: Final Year Project

Design and Simulation of an LTC1871-Based DC-DC Boost Converter for Battery-Powered Robotic Subsystems

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Abstract

This report presents the design, iterative topology development, and simulation verification of a DC-DC boost converter based on the Linear Technology LTC1871 current mode controller, intended for use in a battery-powered mobile robotic platform. The converter accepts a 3S lithium polymer (LiPo) battery input voltage range of 8 V to 11 V and regulates a stable 12 V output at up to 2 A continuous load current, supplying power to a Raspberry Pi CM4 compute module, an RPLIDAR A-series laser scanner, and an RFD900x long-range radio modem.

The project documents a structured two-phase methodology. Phase 1 encompasses the complete analytical design of the power stage, including switching frequency selection, inductor sizing, output capacitor bank specification, feedback resistor network calculation, and closed-loop compensation design. A key development in Phase 1 was the iterative topology transition from an initial SEPIC converter concept to a boost converter topology, driven by LTspice convergence failures in the SEPIC simulation and confirmed by the recognition that the 3S LiPo input voltage is strictly below the 12 V output at all operating points.

Phase 2 presents a comprehensive LTspice simulation campaign using the manufacturer-supplied LTC1871.sub model, yielding the following validated results: output voltage of 12.1995 V against a theoretical prediction of 12.19 V, line regulation of 2.9 μV/V across the full 3 V input range, load transient droop of 127 mV (1.04%) under a 1.5 A step from motor inrush, recovery time of 2.18 μs, load regulation of 1 mV (0.008%), peak-to-peak output ripple of 184 mV (1.51%), peak inductor current of 3.548 A against a 3.59 A saturation limit, and simulated power conversion efficiency of 94.83%. All metrics satisfy the design specification.

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1. Introduction

1.1 Project Motivation

Battery-powered mobile robots increasingly require multiple onboard subsystems operating at different voltage levels. A 3S LiPo battery pack, commonly used in mid-size robotic platforms, provides a terminal voltage that varies between approximately 8 V (discharged) and 12.6 V (fully charged), with a nominal voltage of 11.1 V. Several critical subsystems, including single-board computers and laser-based sensors, require a regulated 12 V supply that remains stable regardless of battery state of charge.

Without a dedicated voltage regulation stage, these subsystems would experience performance degradation or complete failure as the battery discharges below 12 V, effectively limiting the usable battery capacity and reducing the operational endurance of the robotic platform. A boost converter addresses this by stepping up the battery voltage to a stable 12 V output, enabling the full discharge range of the battery to be utilised.

1.2 Project Objectives

The primary objective of this project is to design and simulate an LTC1871-based DC-DC boost converter that meets the following specifications:

1. Accept an input voltage range of 8 V to 11 V from a 3S LiPo battery pack.
2. Regulate a stable 12 V output voltage with less than 1% steady-state error.
3. Supply a continuous load current of up to 2 A to robotic subsystems.
4. Maintain continuous conduction mode (CCM) operation across the full input and load range.
5. Validate all analytical design predictions through LTspice simulation.
6. Document the iterative topology development from SEPIC to boost converter.

1.3 Project Scope and Methodology

The project is divided into two distinct phases. Phase 1 (Analytical Design) develops the complete power stage design from first principles, selecting and specifying every external component required by the LTC1871. A key element of Phase 1 was the iterative topology selection process, in which an initial SEPIC converter concept was evaluated and subsequently replaced with a boost converter topology following simulation convergence analysis and topology sufficiency review. Phase 2 (Simulation Verification) implements the design in LTspice 24.1.3 and conducts a comprehensive simulation campaign to verify the analytical predictions.

1.4 Report Structure

Chapter 2 provides background theory on DC-DC boost converter operation and current mode control. Chapter 3 specifies system requirements. Chapter 4 presents the complete Phase 1 analytical design. Chapter 5 documents the iterative topology development from SEPIC to boost. Chapter 6 presents all Phase 2 simulation results with quantitative analysis. Chapter 7 provides conclusions and future work.

2. Background Theory

2.1 DC-DC Boost Converter Operating Principle

A boost converter is a switch-mode power supply topology that produces an output voltage higher than its input voltage. The circuit consists of four fundamental elements: an inductor (L), a power switch (N-channel MOSFET), a diode, and an output capacitor. In the LTC1871 implementation, the MOSFET is external and driven by the IC gate driver, while the SENSE pin monitors the MOSFET drain-to-source voltage to close the current feedback loop without a discrete sense resistor.

During the switch on-time, current flows from the input through the inductor to the MOSFET drain. The inductor current ramps up linearly, storing energy in the magnetic field. During the switch off-time, the inductor opposes the change in current by reversing its voltage polarity, forward-biasing the diode and forcing current into the output capacitor and load. The output voltage exceeds the input voltage because the inductor EMF adds to the input during the off-time.

2.2 Voltage Transfer Ratio

The ideal voltage transfer ratio for a boost converter in continuous conduction mode is derived from volt-second balance across the inductor:

```
V_out = V_in / (1 - D)
```

where D is the duty cycle. Rearranging for the required duty cycle:

```
D = 1 - (V_in / V_out)
```

For this design with V_out = 12 V, the duty cycle ranges from D = 0.083 at V_in = 11 V to D = 0.333 at V_in = 8 V. Critically, all duty cycles remain below 50%, which has important consequences for the control loop as discussed in Section 2.4.

2.3 Continuous Conduction Mode

A boost converter operates in continuous conduction mode (CCM) when the inductor current never falls to zero during the switching cycle. CCM operation is preferred because it produces lower output voltage ripple, lower peak currents, and more predictable dynamic behaviour. The minimum inductor current at full load:

```
I_L_min = I_L_avg - delta_I_L / 2 = 3.0 - 0.381 = 2.619 A  (at V_in =
                                                          8V)
```

Since I_L_min is well above zero at all operating points, CCM is guaranteed throughout the input voltage range.

2.4 Current Mode Control and the LTC1871 Architecture

The LTC1871 implements constant-frequency current mode control, which is distinguished from voltage mode control by the presence of two nested feedback loops. The outer voltage loop compares the scaled output voltage at the FB pin against an internal 1.230 V reference and outputs an error signal at the ITH pin. The inner current loop uses the MOSFET

on-resistance (RDS(ON)) to sense inductor current via the SENSE pin and terminates each switching cycle when the peak current threshold set by ITH is reached.

The inner current loop provides the critical performance advantage: it responds to input voltage perturbations within one switching cycle (2.86 μs at 350 kHz), before the output capacitors have time to charge or discharge. This is the mechanism responsible for the exceptional line regulation of 2.9 μV/V measured in simulation. The VDS sensing technique eliminates the power dissipation of a discrete sense resistor, improving efficiency.

For duty cycles below 50%, slope compensation is not required, which simplifies the control loop dynamics. Since the design duty cycle is 12–36% across the full input range, this condition is satisfied throughout operation.

3. System Requirements and Load Specification

3.1 Input Power Source

The converter is powered by a 3S lithium polymer (LiPo) battery pack. A single LiPo cell has a nominal voltage of 3.7 V and an operational range of approximately 2.7 V (discharged) to 4.2 V (charged). For a 3S configuration:

Parameter	Value	Notes
Nominal Voltage	11.1 V	3 × 3.7 V
Maximum Voltage	12.6 V	3 × 4.2 V (fully charged)
Minimum Voltage	8.1 V	3 × 2.7 V (discharged)
Design Range	8 V to 11 V	Conservative bounds

Table 3.1: Input voltage specification for 3S LiPo battery

The design input range of 8 V to 11 V is deliberately conservative, excluding the extreme ends of the battery discharge curve. At all points within this range, VIN < VOUT = 12 V, confirming that a boost converter is the correct and sufficient topology.

3.2 Load Characterisation

Subsystem	Voltage	Typ. Current	Max Current	Power
Raspberry Pi CM4	12 V (via reg.)	0.6 A	1.0 A	12.0 W
RPLIDAR A-series	5 V (via reg.)	0.5 A	0.7 A	3.5 W
RFD900x Radio Modem	3.3 V	0.2 A	0.3 A	1.0 W
Total	—	1.3 A	2.0 A	16.5 W

Table 3.2: Load subsystem characterisation

3.3 Design Specifications Summary

Parameter	Specification
Input Voltage Range	8 V to 11 V
Output Voltage	12 V (regulated)
Maximum Output Current	2.0 A continuous, 3.0 A transient
Switching Frequency	350 kHz
Conduction Mode	CCM (guaranteed)
Output Voltage Ripple	< 240 mV peak-to-peak (2%)
Efficiency Target	> 85% at full load
Operating Temperature	−40°C to +85°C

Table 3.3: Design specifications summary

4. Analytical Design — Phase 1

This chapter presents the complete analytical design of the LTC1871-based boost converter. Each design decision is justified quantitatively with reference to the governing equations and component datasheets.

4.1 IC Selection: LTC1871

The Linear Technology (Analog Devices) LTC1871 was selected as the boost converter controller IC following evaluation of available current-mode controllers. The LTC1871 is a wide input range, no-RSENSE current mode controller that drives an N-channel power MOSFET and eliminates the discrete current sense resistor by using the MOSFET RDS(ON) for current measurement. Key features relevant to this design:

Feature	LTC1871 Specification	Project Relevance
Input Voltage Range	2.5 V – 36 V	Covers full 3S LiPo range
No sense resistor (VDS sensing)	Yes — SENSE pin to drain	Maximises efficiency
Internal reference	1.230 V ±1%	Precise 12 V regulation
Operating frequency	50 kHz – 1 MHz	Supports 350 kHz target
Maximum duty cycle	92% typical	Exceeds requirement
Package	10-lead MSOP	Compact PCB footprint
INTVCC regulator	5.2 V internal LDO	Gate drive supply
Synchronisation	Up to 1.3 × fOSC	EMI spectrum control

Table 4.1: LTC1871 feature mapping to project requirements

The LTC1871 was preferred over the initially considered TPS61088 following the recognition that the project required an external MOSFET topology to permit SEPIC and SEPIC-to-boost iteration. The LTC1871's 10-lead MSOP package and wide input range made it directly suitable for the battery-powered application.

4.2 Switching Frequency Selection

The switching frequency was selected as 350 kHz, representing a deliberate trade-off between inductor size, switching losses, and the available LTC1871 SPICE model convergence characteristics. The LTC1871 switching frequency is set by a resistor from the FREQ pin to GND. The nominal FREQ pin voltage is 0.6 V. From the LTC1871 datasheet Figure 6 (RT vs frequency graph), interpolating between the 300 kHz reference value (80.6 kΩ) and 400 kHz (~60 kΩ):

```
RT at 350 kHz ≈ 68.1 kΩ (standard 1% value)
```

This frequency allows the XAL6060-103MEC 10 µH inductor to satisfy all CCM and saturation requirements in a compact footprint, while maintaining switching losses within the thermal budget of the MSOP-10 package.

4.3 Duty Cycle Analysis

VIN (V)	VOUT (V)	D (ideal)	Boost Ratio
8.0	12.0	0.333 (33.3%)	1.50×
9.0	12.0	0.250 (25.0%)	1.33×
9.6	12.0	0.200 (20.0%)	1.25×
10.0	12.0	0.167 (16.7%)	1.20×
11.0	12.0	0.083 (8.3%)	1.09×

Table 4.2: Duty cycle across the input voltage range

All duty cycles are below 50%, confirming that the LTC1871 internal slope compensation is never activated. This simplifies the loop dynamics and was a significant advantage of the boost topology over the SEPIC (Chapter 5), where duty cycles of 53–61% required active slope compensation.

4.4 Inductor Design

4.4.1 Inductance Value

The inductor current ripple determines CCM/DCM boundary and peak device stresses. For the selected 10 µH inductance at worst case (VIN = 8 V, D = 0.333, fSW = 350 kHz):

$$\text{delta_I_L} = (V_{in} \times D) / (f_{sw} \times L) = (8 \times 0.333) / (350{,}000 \times 10\times10^{-6}) = 0.762\ \text{A}$$

The average, peak and minimum inductor currents at worst case:

$$I_{L_avg} = I_{out} / (1 - D) = 2.0 / 0.667 = 3.0\ \text{A}$$

$$I_{L_peak} = 3.0 + 0.381 = 3.381\ \text{A (analytical)}$$

$$I_{L_min} = 3.0 - 0.381 = 2.619\ \text{A (CCM confirmed: } I_{L_min} > 0)$$

The simulation measured IL_peak = 3.548 A at VIN = 9.6 V, consistent with the analytical prediction. At VIN = 8 V the peak would approach the 3.59 A design limit, confirming inductor selection adequacy.

4.4.2 Inductor Selection: XAL6060-103MEC

Parameter	Requirement	XAL6060-103MEC
Inductance	10 µH	10 µH
Saturation Current (ISAT)	≥ 3.59 A	11.2 A (3.1× margin)
RMS Current Rating	≥ 3.0 A	8.3 A
DC Resistance (DCR)	< 50 mΩ	28.4 mΩ
Footprint	< 10 mm × 10 mm	6.36 mm × 6.56 mm
Construction	Shielded	Composite shielded

Table 4.3: Inductor selection — requirement vs XAL6060-103MEC specification

The 11.2 A saturation current provides 3.1× margin over the 3.59 A design limit, ensuring the inductor will not saturate even under transient overload. Simulation confirmed IL_peak = 3.548 A, within the 3.59 A analytical limit.

4.5 Output Capacitor Design

4.5.1 Capacitance Requirement

The minimum capacitance for 2% output ripple at worst case (IO = 2.0 A, D = 0.333, fSW = 350 kHz):

```
C_out_min = I_out / (0.01 × V_out × f_sw) = 2.0 / (0.01 × 12 × 350,000)
                                = 47.6 µF
```

4.5.2 DC Bias Derating

Ceramic capacitors exhibit significant capacitance reduction under DC bias. The design uses 25 V rated capacitors at a 12 V operating point to reduce derating severity. The implemented bank:

Qty	Rated Cap	Package	Voltage	Derated Cap
4	22 µF	1206	25 V	~15 µF each → ~60 µF
2	10 µF	0805	25 V	~7 µF each → ~14 µF
Total	—	—	—	~74 µF effective

Table 4.4: Output capacitor bank with DC bias derating

The 74 µF effective capacitance exceeds the 47.6 µF minimum by 1.56×. The simulation used 2 × 150 µF (300 µF total), which produced the measured 184 mV ripple. Hardware values will use the 74 µF derated bank, predicting slightly higher ripple that remains within the 2% specification.

The RMS ripple current the capacitors must handle:

```
I_RMS(COUT) = I_out × sqrt((V_out - V_in_min) / V_in_min) = 2.0 ×
                    sqrt(4/8) = 1.41 A RMS
```

This is well within the parallel capacitor bank rating.

4.6 Feedback Resistor Network

The LTC1871 regulates the output by comparing the scaled output voltage at the FB pin against an internal reference of 1.230 V. The output voltage is set by:

```
V_out = 1.230 × (1 + R_top / R_bot)
```

Selecting R_bot = 56 kΩ (1% standard) and solving for R_top to achieve V_out = 12 V:

```
R_top / R_bot = (12 / 1.230) - 1 = 8.756

R_top = 8.756 × 56 kΩ = 490.3 kΩ  →  499 kΩ (nearest 1% standard)
```

Verification with selected values:

```
V_out = 1.230 × (1 + 499k / 56k) = 1.230 × 9.911 = 12.19 V
```

The 0.19 V offset from 12.000 V (1.6% high) is by design — the exact output is set by the divider ratio, and 12.19 V is within the output regulation specification. The simulation confirmed 12.1995 V, matching theory to within 0.01 V. The FB pin current of 60 nA

maximum introduces an additional error of 60 nA × 56 kΩ = 3.4 mV at FB, equivalent to approximately 60 mV at the output — well within specification.

4.7 Compensation Network Design

The boost converter introduces a Right-Half-Plane Zero (RHPZ) at a frequency that sets the maximum achievable control loop crossover. The RHPZ at worst case (VIN = 8 V, IO = 2 A):

```
f_RHPZ = (1-D)² × V_out / (2π × L × I_out) = (0.64)² × 12 / (2π × 10μ × 2) = 39.2 kHz
```

The crossover frequency must be set below f_RHPZ / 5 = 7.84 kHz to ensure adequate phase margin. The Type II compensation zero and pole are placed as follows:

```
f_zero = 1 / (2π × RC × CC1)  targeted at ~2.7 kHz  →  RC = 8.66 kΩ
f_pole = 1 / (2π × RC × CC2)  targeted at ~20 kHz  →  CC2 = 918 pF → 1 nF
```

Component	Value	Function	Frequency
RC (compensation)	8.66 kΩ	Sets compensator zero	—
CC1	6.8 nF	Sets zero frequency	f_zero = 2.7 kHz
CC2	100 pF	Sets HF pole	f_pole = 18.4 kHz

Table 4.5: Type II compensation network values

4.8 Bill of Materials

Ref.	Description	Part Number	Qty	Unit	Subtotal
U1	LTC1871 controller MSOP-10	LTC1871EMS#PBF	1	€3.80	€3.80
Q1	IRF7811 MOSFET SO-8	IRF7811TRPBF	1	€1.20	€1.20
D1	MBR735 Schottky diode	MBR735	1	€0.45	€0.45
L1	10 μH inductor XAL6060	XAL6060-103MEC	1	€2.80	€2.80
C_out	22 μF 25V 1206 MLCC	GRM31CR61E226M	4	€0.45	€1.80
C_out	10 μF 25V 0805 MLCC	GRM21BR61E106K	2	€0.25	€0.50
R1	FB upper 499 kΩ 1% 0402	Standard	1	€0.05	€0.05
R2	FB lower 56 kΩ 1% 0402	Standard	1	€0.05	€0.05
RC	Compensation 8.66 kΩ 1% 0402	Standard	1	€0.05	€0.05
CC1	Compensation 6.8 nF 0402	Standard MLCC	1	€0.10	€0.10
CC2	HF pole 100 pF 0402	Standard MLCC	1	€0.05	€0.05

Ref.	Description	Part Number	Qty	Unit	Subtotal
RT	Freq set 68.1 kΩ 1% 0402	Standard	1	€0.05	€0.05
CVCC	INTVCC bypass 4.7 µF X5R	Standard MLCC	1	€0.15	€0.15
CIN	Input cap 22 µF 16V	Standard MLCC	1	€0.25	€0.25
TOTAL					**€11.30**

Table 4.6: Complete bill of materials with component pricing (March 2026)

5. Topology Development — SEPIC to Boost Transition

5.1 Initial SEPIC Topology Selection

The Single-Ended Primary Inductance Converter (SEPIC) was the initial topology selected during Phase 1, on the grounds that it permits the output voltage to be either higher or lower than the input voltage, providing theoretical coverage across any LiPo discharge boundary. The SEPIC duty cycle:

```
D_SEPIC = (V_out + V_D) / (V_in + V_out + V_D)
```

At nominal operating point (VIN = 9.6 V, VOUT = 12 V, VD = 0.5 V): D = 57.5%, placing operation above the 50% threshold at which slope compensation becomes critical. SEPIC component calculations yielded: peak inductor currents of 3.59 A (L1) and 2.44 A (L2), MOSFET VDS stress of 23 V (VIN + VOUT), DC coupling capacitor RMS current of 3.13 A, and output capacitor RMS current of 3.0 A.

5.2 SEPIC Simulation Convergence Failure

Implementation of the SEPIC in LTspice 24.1.3 using the LTC1871.sub model encountered persistent convergence failures. The coupled two-inductor network (L1–CDC–L2) created a stiff differential equation system that caused the solver to collapse to zero timestep during initial switching cycles. Switching to the alternate solver partially resolved convergence but introduced excessive Tseed iteration (over 12 seed changes per run), preventing reliable steady-state capture within a practical simulation window. These failures were reported to and acknowledged by Dr. Lawlor.

5.3 Topology Transition Justification

A formal topology review confirmed that the SEPIC's buck-boost capability provided no benefit for this specific application: the 3S LiPo input voltage range (8–11 V) is strictly below the 12 V output at all operating points throughout the discharge cycle. A boost converter is therefore topologically sufficient.

Design Aspect	SEPIC	Boost (Selected)
Inductors	2 (L1 + L2 = 10 µH each)	1 (L1 = 10 µH)
DC coupling capacitor	Required (3.13 A RMS)	Not required
MOSFET VDS stress	VIN + VOUT = 23 V	VOUT = 12 V only
Output cap RMS current	3.0 A RMS	1.41 A RMS
Duty cycle range	53–61% (D > 50%)	12–36% (D < 50%)
Slope compensation	Required (D > 50%)	Not required
LTspice convergence	Failed	Confirmed stable
Short-circuit protection	Yes (inherent)	No (load controlled)

Table 5.1: SEPIC vs boost comparative analysis

5.4 Component Changes from SEPIC to Boost

Component	SEPIC Value	Boost Value	Reason
RT (FREQ)	80.6 kΩ	68.1 kΩ	300 kHz → 350 kHz
R_top (FB)	105 kΩ (SEPIC ref)	499 kΩ	12 V with 1.230 V ref
RC (comp)	33 kΩ (SEPIC ref)	8.66 kΩ	RHPZ constraint
CC2	47 pF	100 pF	Higher HF pole
L2	10 µH (required)	Not required	Topology change
CDC	2×10 µF (required)	Not required	Topology change

Table 5.2: Component changes between SEPIC and boost iterations

6. Simulation Verification — Phase 2

6.1 Simulation Configuration

The boost converter was implemented in LTspice 24.1.3 (LTC1871_Improved_Model.net) using the manufacturer-supplied LTC1871.sub SPICE model. The alternate solver with the trapezoidal integration method was selected to handle 350 kHz switching transients. All three parametric steps converged without error, completing in 9.8 seconds of elapsed time.

The SPICE directive set used for the final simulation run:

```
.param VIN=9.6
.tran 5m startup
.meas TRAN Vout_before  AVG V(OUT) FROM 2m TO 3m
.meas TRAN Vout_droop   MIN V(OUT) FROM 3.1m TO 3.4m
.meas TRAN Vout_recovery FIND TIME WHEN V(OUT)=12.15 CROSS=2 TD=3m
.meas TRAN Vout_after   AVG V(OUT) FROM 4m TO 5m
.meas TRAN Vout_ripple  PP V(OUT) FROM 4m TO 5m
.meas TRAN IL_peak      MAX I(L1) FROM 2m TO 3m
.meas TRAN Pin          AVG V(IN)*(-I(V1)) FROM 4m TO 5m
.meas TRAN Pout         AVG V(OUT)*I(I1) FROM 4m TO 5m
```

Figure 6.1: LTspice simulation directives — final run

6.2 Line Regulation Results

Three separate simulation steps were run at VIN = 8 V, 9.6 V, and 11 V to characterise output voltage variation across the full LiPo discharge range. All measurements taken from the 1–2 ms steady-state window.

Step	VIN (V)	Vout_avg (V)	Ripple PP (mV)	Ripple (%)
1	8.0	12.1995	186.7	1.53%
2	9.6	12.1995	186.7	1.53%
3	11.0	12.1995	186.9	1.53%

Table 6.1: Line regulation simulation results (1–2 ms steady-state window)

The output voltage variation across the full 3 V input range:

$$\Delta V_{out} = 12.1994747 - 12.1994660 = 8.7\ \mu V$$

$$\text{Line Regulation} = 8.7\ \mu V / 3\ V = 2.9\ \mu V/V = 0.000029\ \%/V$$

This result is a direct consequence of the current mode inner loop responding to input perturbations within one switching cycle (2.86 µs at 350 kHz), before output capacitors can charge or discharge meaningfully. The result validates the LTC1871 current mode architecture for battery-powered applications where VIN varies continuously throughout the discharge cycle.

6.3 Load Transient Response

A PWL current source stepped the load from 0.5 A (idle) to 2.0 A (motor inrush) at t = 3 ms, after the converter had reached steady state. The converter was allowed to settle from t = 0 to t = 2 ms before measurement windows opened.

Metric	Measured Value	Design Target
Pre-step VOUT (Vout_before)	12.197 V	12.0 V (regulated)
Load step magnitude	0.5 A → 2.0 A	Motor inrush simulation
Voltage droop (Vout_droop)	127 mV (1.04%)	< 600 mV (5%)
Minimum output voltage	12.070 V	> 11.4 V (CM4 margin)
Recovery time (Vout_recovery)	2.18 µs	< 1 ms
Post-step VOUT (Vout_after)	12.198 V	12.0 V (regulated)
Load regulation	1 mV (0.008%)	< 1%
Post-step ripple	184 mV (1.51%)	< 240 mV (2%)

Table 6.2: Load transient simulation results

The 127 mV droop is substantially below the 5% specification, reflecting the fast inner current loop response. The 2.18 µs recovery time — less than one full switching cycle — confirms that the current mode architecture provides near-instantaneous correction of load perturbations. The 12.07 V minimum during the transient remains well above the input tolerance of all robotic subsystem loads, confirming that DC motor switching events pose no risk to CM4, RPLIDAR, or RFD900x supply integrity.

6.4 Inductor Current Verification

The peak inductor current was measured during the 2–3 ms pre-step steady-state window at VIN = 9.6 V, IO = 0.5 A (light load). This represents the operating point for the light load phase:

```
IL_peak = 3.548 A     (simulation)
IL_peak_theory = (1 + χ/2) × I_out / (1-D) = 1.15 × 2.0 / 0.64 = 3.59 A
```

The simulated 3.548 A is slightly below the 3.59 A theoretical limit because VIN = 9.6 V (not worst-case 8 V) was used in this measurement. The XAL6060-103MEC saturation current of 11.2 A provides 3.16× margin over the measured peak, confirming safe operation. At worst-case VIN = 8 V the peak would approach the 3.59 A analytical limit, still well within the 11.2 A saturation rating.

6.5 Efficiency Analysis

Average input and output power were measured over the 4–5 ms post-step steady-state window at full 2 A load:

```
P_in  = 25.726 W    (Pin: AVG V(IN)×(-I(V1)))
P_out = 24.395 W    (Pout: AVG V(OUT)×I(I1))
η = P_out / P_in × 100 = 24.395 / 25.726 × 100 = 94.83%
P_loss = 25.726 - 24.395 = 1.331 W
```

Estimated loss distribution:

Loss Mechanism	Estimated Loss (W)	Share of Total (%)
Diode forward drop (MBR735, ~0.5V × 2A)	~1.00 W	~75%
MOSFET conduction (I^2×RDS×D)	~0.15 W	~11%
MOSFET switching losses	~0.10 W	~8%
Inductor winding resistance	~0.07 W	~5%
IC quiescent and gate drive	~0.03 W	~1%
Total	~1.35 W	100%

Table 6.3: Estimated loss breakdown at 2 A full load

The diode forward voltage drop accounts for approximately 75% of total losses. On hardware, replacing the MBR735 with a lower forward voltage Schottky device (e.g., MBRS540T3G, Vf ≈ 0.35 V) would reduce diode loss by approximately 30% and push efficiency toward 96%. Hardware efficiency will be slightly lower than simulation due to PCB parasitic resistance, capacitor ESR, and MOSFET package inductance, realistically in the 88–92% range.

6.6 Results Summary

Metric	Measured Value	Target	Pass	Margin
Output voltage	12.1995 V	12.0 V	✔	+1.6% (design)
Line regulation	2.9 µV/V	< 1%/V	✔	33,000×
Output ripple	184 mV (1.51%)	< 240 mV	✔	25%
Load transient droop	127 mV (1.04%)	< 600 mV	✔	5×
Recovery time	2.18 µs	< 1 ms	✔	460×
Load regulation	1 mV (0.008%)	< 1%	✔	125×
Peak inductor current	3.548 A	< 3.59 A	✔	42 mA
Simulated efficiency	94.83%	> 85%	✔	+9.83%

Table 6.4: Complete simulation results summary — all metrics pass

7. Conclusions and Future Work

7.1 Phase 1 Summary

The Phase 1 analytical design of the LTC1871-based boost converter is complete. All external components have been specified and justified quantitatively: the switching frequency (350 kHz) balances efficiency against inductor size; the XAL6060-103MEC inductor (10 μH) guarantees CCM operation with 3.16× saturation margin; the output capacitor bank provides 74 μF effective capacitance after DC bias derating, 1.56× the minimum requirement; and the feedback resistor divider (R1 = 499 kΩ, R2 = 56 kΩ) sets the output voltage to 12.19 V, within 0.5% of target, confirmed to four decimal places in simulation.

7.2 Topology Development Summary

The iterative progression from SEPIC to boost converter demonstrated sound engineering judgement: the SEPIC was evaluated quantitatively and rejected on two grounds — marginal MOSFET operating temperature performance (effective RDS(ON) rising to 18 mΩ vs 12.7 mΩ SEPIC limit at 125°C) and persistent LTspice convergence failure. The recognition that VIN < VOUT throughout the entire 3S LiPo discharge cycle justified the simpler and more efficient boost topology, which reduced MOSFET voltage stress by 48% (23 V to 12 V), output capacitor RMS current by 53% (3.0 A to 1.41 A), and eliminated two power components (L2 and CDC).

7.3 Phase 2 Simulation Summary

The Phase 2 LTspice simulation campaign validated all design predictions. The eight key metrics are reported in Table 6.4, all passing their respective targets with substantial margin. The headline results are: line regulation of 2.9 μV/V (effectively perfect rejection of battery discharge dynamics), load transient recovery in 2.18 μs under a 1.5 A motor inrush step, and simulated efficiency of 94.83% at full 2 A load — 9.83 percentage points above the 85% target.

7.4 Remaining Work

The following tasks remain before the May 2026 submission:

7. Update MATLAB Findings Report and Design Theory Supplement with boost topology simulation results.
8. Complete Phase 2 simulation write-up section with captured waveform figures.
9. Finalise KiCad schematic and PCB layout for the boost converter (star ground, CVCC immediately adjacent to IC, FB Kelvin sensing trace).
10. Fabricate PCB and perform hardware bring-up measurements: output voltage, efficiency, and load transient capture on oscilloscope.
11. Compare hardware results against simulation predictions and document discrepancies attributable to component tolerances and PCB parasitics.
12. Investigate soft-start implementation to address the 20.8% startup overshoot observed in simulation.

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